# Switch-mode Power Rectifiers MBR560MFS, NRVB560MFS

## SCHOTTKY BARRIER RECTIFIERS 5 AMPERES 60 VOLTS



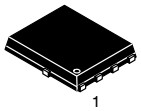

These state-of-the-art devices have the following features:

### Features

- Low Power Loss / High Efficiency
- New Package Provides Capability of Inspection and Probe After Board Mounting
- Guardring for Stress Protection
- Low Forward Voltage Drop
- 175 °C Operating Junction Temperature
- NRVB Prefix for Automotive and Other Applications Requiring Unique Site and Control Change Requirements; AEC-Q101 Qualified and PPAP Capable
- Wettable Flanks Option available
- These are Pb-Free Devices

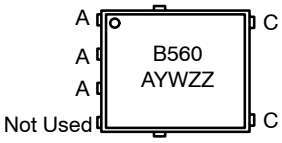
### Mechanical Characteristics:

- Case: Epoxy, Molded
- Epoxy Meets Flammability Rating UL 94−0 @ 0.125 in.
- Lead Finish: 100% Matte Sn (Tin)
- Lead and Mounting Surface Temperature for Soldering Purposes: 260°C Max. for 10 Seconds
- Device Meets MSL 1 Requirements
- The Heat Generated must be less than the Thermal Conductivity from Junction−to−Ambient: dPD/dTJ < 1/RJA




**SO−8 FLAT LEAD CASE 488AA STYLE 2**

**MARKING DIAGRAM**




| | |
|---|---|
| B560 | = Specific Device Code |
| A | = Assembly Location |
| Y | = Year |
| W | = Work Week |
| ZZ | = Lot Traceability |

### MAXIMUM RATINGS

| Rating | Symbol | Value | Unit |
|---|---|---|---|
| Peak Repetitive Reverse Voltage<br>Working Peak Reverse Voltage<br>DC Blocking Voltage | $V_{RRM}$<br>$V_{RWM}$<br>$V_R$ | 60 | V |
| Average Rectified Forward Current (Rated $V_R$, $T_C$ = 160 °C) | $I_{F(AV)}$ | 5.0 | A |
| Peak Repetitive Forward Current, (Rated $V_R$, Square Wave, 20 kHz, $T_C$ = 165 °C) | $I_{FRM}$ | 10 | A |
| Non−Repetitive Peak Surge Current (Surge Applied at Rated Load Conditions Halfwave, Single Phase, 60 Hz) | $I_{FSM}$ | 100 | A |
| Storage Temperature Range | $T_{stg}$ | −65 to +175 | °C |
| Operating Junction Temperature | $T_J$ | −55 to +175 | °C |
| Unclamped Inductive Switching Energy (10 mH Inductor, Non−repetitive) | $E_{AS}$ | 10 | mJ |
| ESD Rating (Human Body Model) | | 3B | |
| ESD Rating (Machine Model) | | M4 | |

Stresses exceeding Maximum Ratings may damage the device. Maximum Ratings are stress ratings only. Functional operation above the Recommended Operating Conditions is not implied. Extended exposure to stresses above the Recommended Operating Conditions may affect device reliability.

### ORDERING INFORMATION

| Device | Package | Shipping† |
|---|---|---|
| NRVB560MFST1G | SO−8 FL (Pb−Free) | 1500 / Tape & Reel |
| NRVB560MFST3G | SO−8 FL (Pb−Free) | 5000 / Tape & Reel |

**DISCONTINUED** (Note 1)

| | | |
|---|---|---|
| MBR560MFST1G | TO−220 (Pb−Free) | 1500 / Tape & Reel |
| MBR560MFST3G | TO−220 (Pb−Free) | 5000 / Tape & Reel |

†For information on tape and reel specifications, including part orientation and tape sizes, please refer to our Tape and Reel Packaging Specifications Brochure, BRD8011/D.

1. **DISCONTINUED:** This device is not recommended for new design. Please contact your **onsemi** representative for information. The most current information on this device may be available on www.onsemi.com.

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## THERMAL CHARACTERISTICS

| Characteristic | Symbol | Typ | Max | Unit |
|---|---|---|---|---|
| Thermal Resistance, Junction-to-Case, Steady State<br>(Assumes 600 $mm^2$ 1 oz. copper bond pad, on a FR4 board) | $R_{\theta JC}$ | – | 2.4 | °C/W |

## ELECTRICAL CHARACTERISTICS

| | | | | |
|---|---|---|---|---|
| Instantaneous Forward Voltage (Note 1)<br>($i_F$ = 5 Amps, $T_J$ = 100°C)<br>($i_F$ = 5 Amps, $T_J$ = 25°C) | $v_F$ | <br>0.60<br>0.65 | <br>0.72<br>0.78 | V |
| Instantaneous Reverse Current (Note 1)<br>(Rated dc Voltage, $T_J$ = 100°C)<br>(Rated dc Voltage, $T_J$ = 25°C) | $i_R$ | <br>1.5<br>0.015 | <br>2.5<br>0.150 | mA |

1. Pulse Test: Pulse Width = 300 μs, Duty Cycle ≤ 2.0%.



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## TYPICAL CHARACTERISTICS

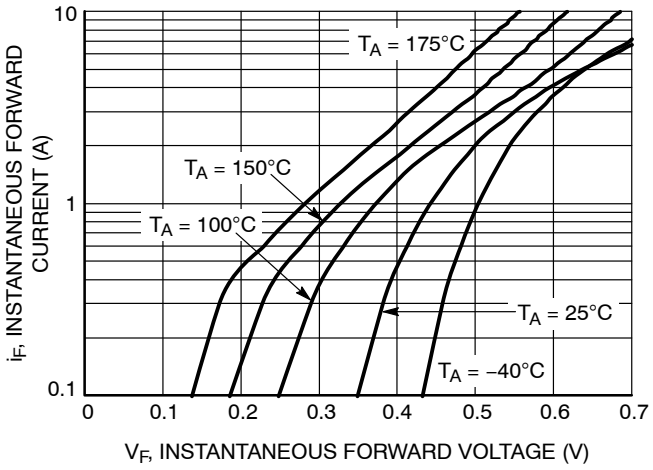



**Figure 1. Typical Instantaneous Forward Characteristics**

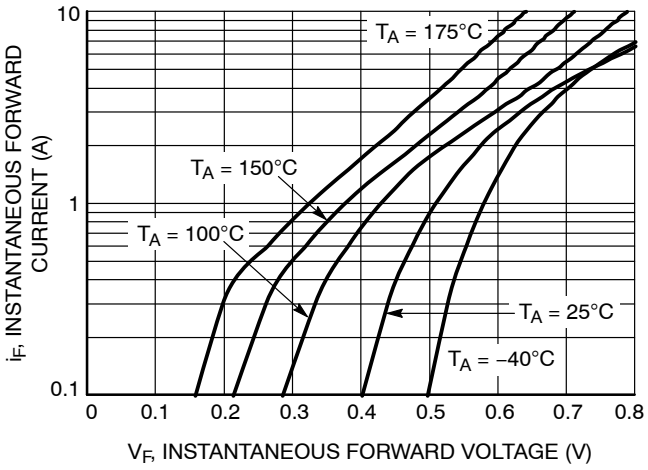



**Figure 2. Maximum Instantaneous Forward Characteristics**

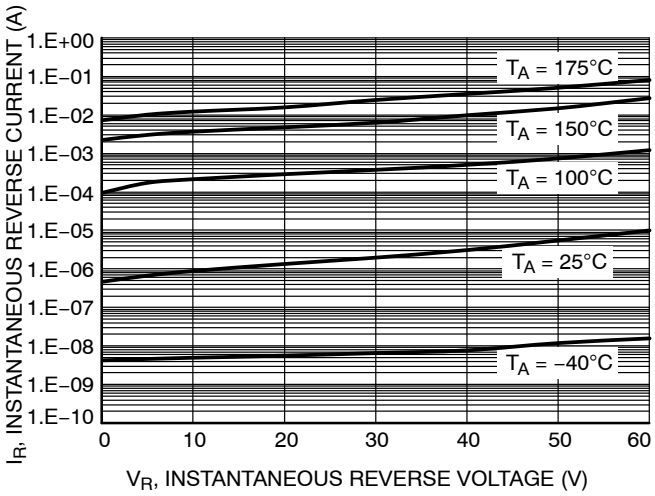



**Figure 3. Typical Reverse Characteristics**

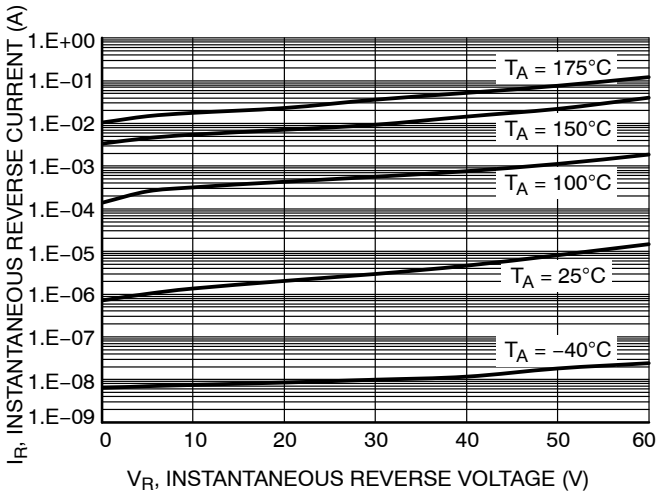



**Figure 4. Maximum Reverse Characteristics**

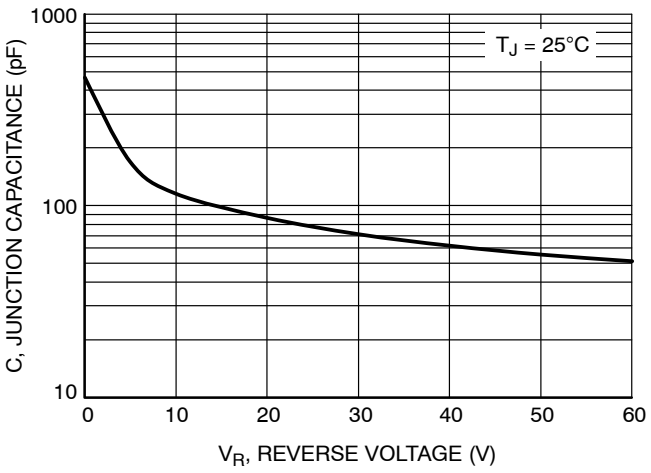



**Figure 5. Typical Junction Capacitance**

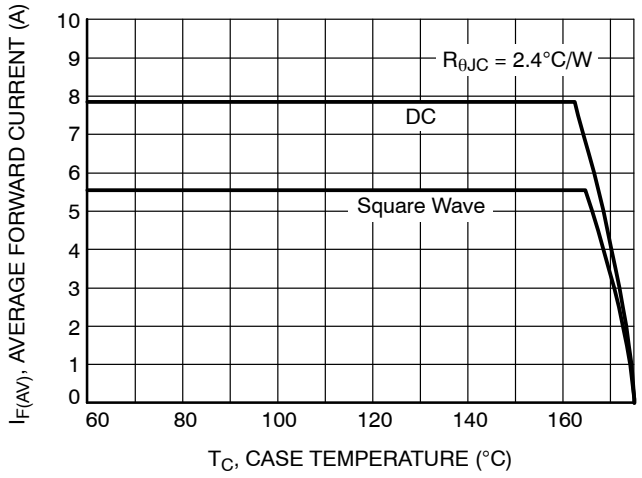



**Figure 6. Current Derating TO−220AB**



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## TYPICAL CHARACTERISTICS

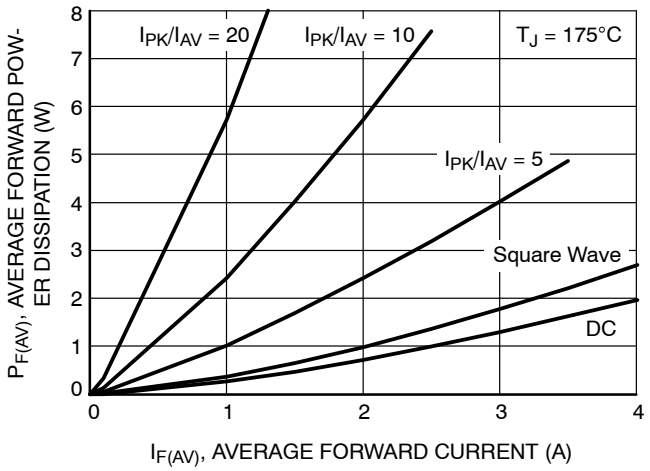



**Figure 7. Forward Power Dissipation**

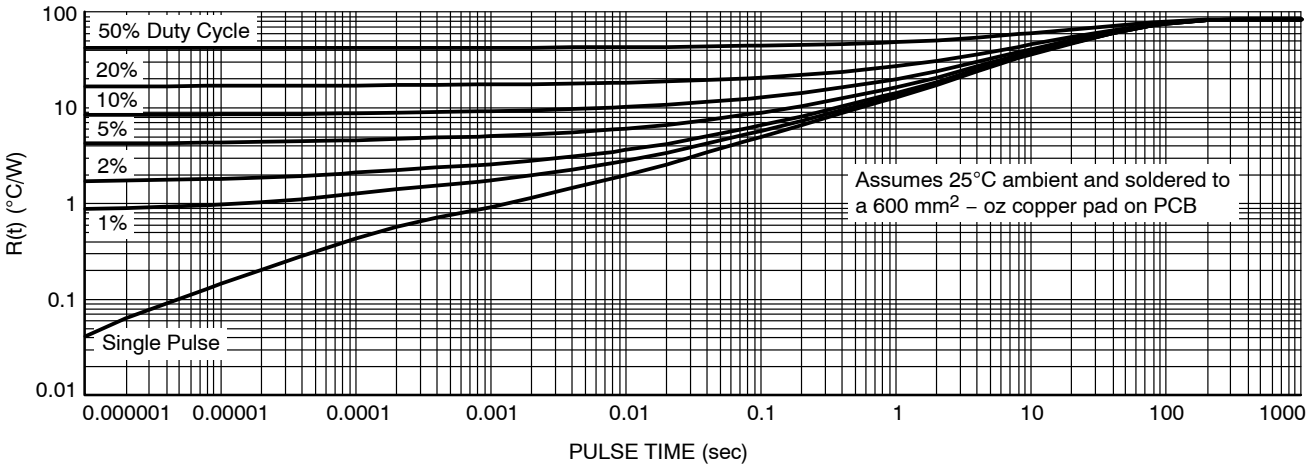



**Figure 8. Thermal Characteristics**



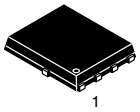
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## REVISION HISTORY

| Revision | Description of Changes | Date |
|---|---|---|
| 2 | MBR560MFST1G, MBR560MFST3G OPN Marked as Discontinued + Rebranded the Data Sheet to onsemi format | 6/25/2025 |



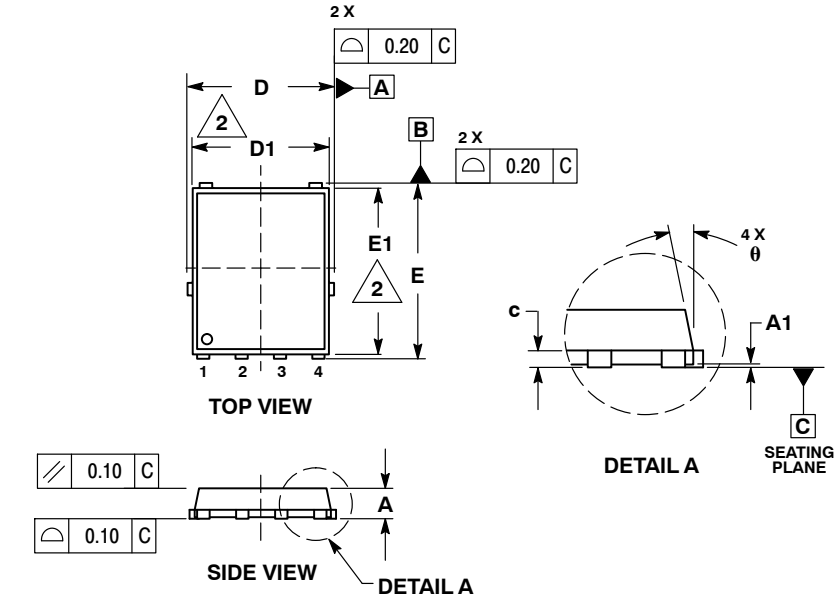
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**SCALE 2:1**

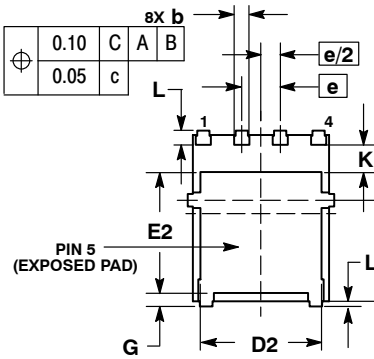
**DFN5 5x6, 1.27P**
**(SO−8FL)**
CASE 488AA
ISSUE N

DATE 25 JUN 2018

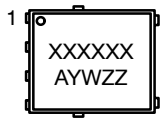



NOTES:
1. DIMENSIONING AND TOLERANCING PER ASME Y14.5M, 1994.
2. CONTROLLING DIMENSION: MILLIMETER.
3. DIMENSION D1 AND E1 DO NOT INCLUDE MOLD FLASH PROTRUSIONS OR GATE BURRS.

| | MILLIMETERS | | |
|---|---|---|---|
| DIM | MIN | NOM | MAX |
| A | 0.90 | 1.00 | 1.10 |
| A1 | 0.00 | ––– | 0.05 |
| b | 0.33 | 0.41 | 0.51 |
| c | 0.23 | 0.28 | 0.33 |
| D | 5.00 | 5.15 | 5.30 |
| D1 | 4.70 | 4.90 | 5.10 |
| D2 | 3.80 | 4.00 | 4.20 |
| E | 6.00 | 6.15 | 6.30 |
| E1 | 5.70 | 5.90 | 6.10 |
| E2 | 3.45 | 3.65 | 3.85 |
| e | 1.27 BSC | | |
| G | 0.51 | 0.575 | 0.71 |
| K | 1.20 | 1.35 | 1.50 |
| L | 0.51 | 0.575 | 0.71 |
| L1 | 0.125 REF | | |
| M | 3.00 | 3.40 | 3.80 |
| θ | 0 ° | ––– | 12 ° |



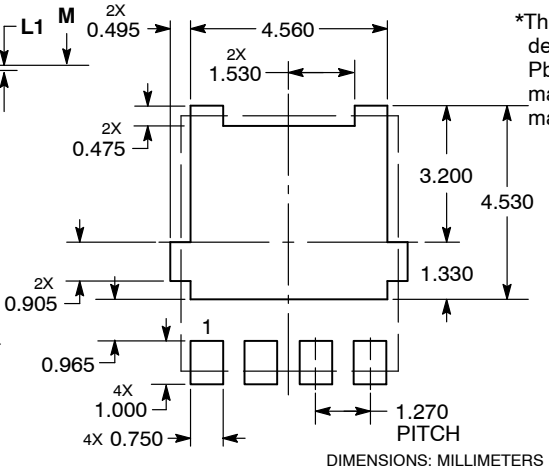

**GENERIC MARKING DIAGRAM***




XXXXXX = Specific Device Code
A = Assembly Location
Y = Year
W = Work Week
ZZ = Lot Traceability

*This information is generic. Please refer to device data sheet for actual part marking. Pb−Free indicator, “G” or microdot “ ▪”, may or may not be present. Some products may not follow the Generic Marking.

**RECOMMENDED SOLDERING FOOTPRINT***




DIMENSIONS: MILLIMETERS

STYLE 1:
PIN 1. SOURCE
2. SOURCE
3. SOURCE
4. GATE
5. DRAIN

STYLE 2:
PIN 1. ANODE
2. ANODE
3. ANODE
4. NO CONNECT
5. CATHODE

*For additional information on our Pb−Free strategy and soldering details, please download the **onsemi** Soldering and Mounting Techniques Reference Manual, SOLDERRM/D.

| | | | |
|---|---|---|---|
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| **DESCRIPTION:** | **DFN5 5x6, 1.27P (SO−8FL)** | | **PAGE 1 OF 1** |

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